



Python Programming - 2301CS404

Lab - 11

Roll No : 514

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Progrmas based on Python Data Structure + UDF + File Handling

Practical 1: Count Lines, Words and Characters

Problem Definition:

Write a Python program to read text from a file and count the total number of lines, words, and characters. The result should be written to an output file.

```
In [2]: f5 = open("newFile.txt",'r')
f6 = open("newFileResult.txt",'a')
wordCount = 0
lineCount = 0
characterCount = 0
for i in f5.read():
    characterCount += 1
f5.close()

f5 = open("untitled.txt",'r')
for i in f5.readlines():
    lineCount += 1
f5.close()

f5 = open("untitled.txt",'r')
for i in f5.readline():
    wordCount+= 1

print("Character Count:",characterCount)
print("Line Count:",lineCount)
```

```
print("Word Count:", wordCount)
f5.close()
```

Character Count: 93

Line Count: 6

Word Count: 13

Practical 2: Search a Word in a File

Problem Definition:

Write a Python program to search a given word in a text file and count its occurrences. Store the result in an output file.

In []:

```
In [5]: word = input("Enter the word to search: ")
fp1 = open("untitled.txt", "r")

wordCount = 0
for i in fp1:
    words = i.split()
    wordCount += words.count(word)
fp1.close()

fp2 = open("output2.txt", "w")
fp2.write(word + " occurred " + str(wordCount) + " times.")
fp2.close()
```

Practical 3: Student Marks and Average

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file. Store marks in a list, calculate the average using a user-defined function, and write student details along with average marks to an output file.

In []:

```
In [2]: def average(marks):
        return sum(marks) / len(marks)

fp1 = open("input.txt", 'r')
fp2 = open("result.txt", "a")

for i in fp1:
    li = i.split()
    rollno = li[0]
    name = li[1]
    marks = li[2:]
    marks = list(map(int, marks))
    avg = average(marks)
    print(avg)
    fp2.write(name + " " + str(avg) + "\n")

fp1.close()
fp2.close()
```

80.0
28.0
52.0
42.0
50.0

Practical 4: Maximum and Minimum using Tuple

Problem Definition:

Write a Python program to read numbers from a file, store them in a tuple, and find the maximum and minimum values. Write the result to an output file.

```
In [23]: fp1 = open("input4.txt", "r")
fp2 = open("output4.txt", "w")
tup = fp1.read().split()
num = tuple(map(int, tup))

if numbers:
    maximum = max(num)
    minimum = min(num)
    fp2.write("Maximum Value: " + str(maximum) + "\n")
    fp2.write("Minimum Value: " + str(minimum) + "\n")
```

Practical 5: Remove Duplicate Words using Set

Problem Definition:

Write a Python program to read words from a file and remove duplicate words using a set. Store the unique words in an output file.

```
In [25]: fp1 = open("untitled.txt", "r")
data = fp1.read()
words = data.split()
distinct = set(words)

fp2 = open("output5.txt", "w")

for word in distinct:
    fp2.write(word + " ")

fp1.close()
fp2.close()
```

Practical 6: Word Frequency using Dictionary

Problem Definition:

Write a Python program to count the frequency of each word in a text file using a dictionary and write the result to an output file.

```
In [27]: fp1 = open("untitled.txt", "r")
data = fp1.read()

words = data.lower().split()
freq = {}
```

```

for word in words:
    if word in freq:
        freq[word] += 1
    else:
        freq[word] = 1

fp2 = open("output6.txt", "w")

for word in freq:
    fp2.write(word + " : " + str(freq[word]) + "\n")

fp1.close()
fp2.close()

```

Practical 7: Student Result and Grade Calculation

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file. Use a user-defined function to determine grade (A/B/C/Fail) and write the student result to an output file.

Grade Criteria:

- Marks ≥ 75 : Grade A
- Marks ≥ 60 : Grade B
- Marks ≥ 40 : Grade C
- Marks < 40 : Fail

```

In [1]: def grade(marks):
        if marks >= 75:
            return "A"
        elif marks >= 60:
            return "B"
        elif marks >= 40:
            return "C"
        else:
            return "Fail"

        fp1 = open("input.txt", "r")
        fp2 = open("output7.txt", "w")

        for line in fp1:
            data = line.strip().split()
            rollno = data[0]
            name = data[1]
            marks = int(data[2])
            grades = grade(marks)
            fp2.write(rollno + " " + name + " " + str(marks) + " " + grades + "\n")

        fp1.close()
        fp2.close()

```

In []: